

Reference Image 4 — Blood Supply of the Brain

The map every stroke question is drawn on: territory predicts the deficit, and vessel calibre predicts the lesion

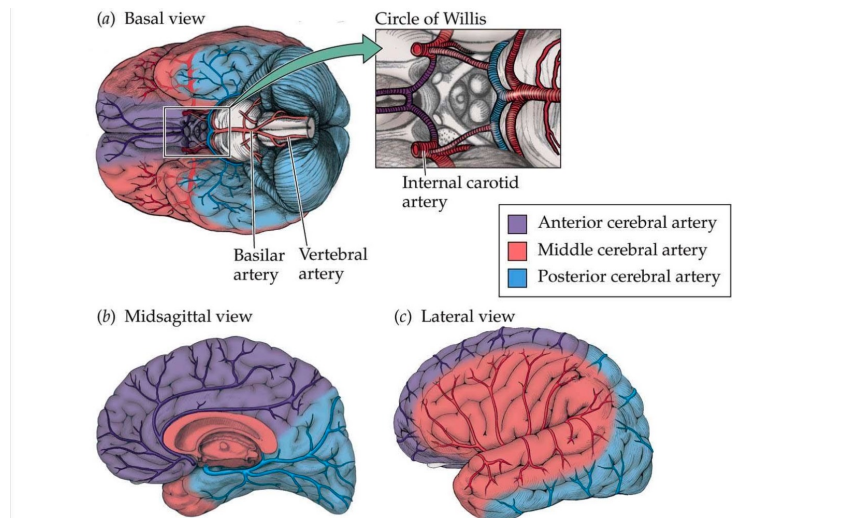


Figure 1. (a) Basal view with the circle of Willis inset; (b) midsagittal and (c) lateral views. **ACA purple, MCA red, PCA blue.**

Table 1. Territory maps to deficit

Artery	Territory	Classic deficit
ACA	Medial frontal, paracentral lobule	Contralateral leg > arm ; abulia
MCA	Lateral convexity; lenticulostrates	Face/arm > leg ; aphasia or neglect
PCA	Occipital, medial temporal	Homonymous hemianopia , macular sparing
Lenticulostriate	Putamen, internal capsule	Lacunar stroke; hypertensive ICH

IDENTIFY: Three views of the cerebral arterial supply, colour-coded by territory, with the **circle of Willis** enlarged from the basal view and the **internal carotid, vertebral and basilar** arteries labelled.

TWO CIRCULATIONS: **Anterior** — internal carotid → ACA + MCA (plus ophthalmic, posterior communicating, anterior choroidal); supplies most of the cerebral hemispheres. **Posterior** — vertebral → basilar → PCA, with PICA, AICA and SCA; supplies brainstem, cerebellum and occipital lobes.

CIRCLE OF WILLIS: Anterior communicating joins the two ACAs; posterior communicating joins the ICA to the PCA. **It is a collateral reserve** — a slowly occluding ICA can be silent, while a sudden embolus is not. It is also where berry aneurysms sit; the **anterior communicating artery is the most common site**; a **PCom aneurysm compresses CN III** → blown pupil with a down-and-out eye.

LARGE vs SMALL VESSELS: Large surface arteries occlude by **atherosclerosis at branch points** (ICA bifurcation, MCA origin, basilar) or by embolus → wedge-shaped cortical infarcts. The **lenticulostriate penetrators** are small, end-arterial and without collaterals; chronic hypertension makes them **lipohyalinotic** → **lacunar infarcts and hypertensive hemorrhage in the same deep territory**.

WATERSHED ZONES: The ACA–MCA and MCA–PCA border zones sit at the end of two supplies. Global hypoperfusion (arrest, sepsis, cardiac surgery) infarcts them first → **proximal arm and leg weakness with the face spared** (“man in a barrel”).

SELECTIVE VULNERABILITY: Ranked by sensitivity to hypoxia: **hippocampus**, cerebellar Purkinje cells, and neocortical layers 3, 5 and 6 (laminar necrosis).

III → cortex

V → brainstem/spinal cord

VI → thalamus

WHY IT MATTERS: **occlusion of a vessel (infarction) and rupture of one (hemorrhage)**

Slide 5 — Cerebral Infarction

Liquefactive necrosis in a vascular territory — the gross appearance dates the lesion



EVOLUTION — GROSS AND MICRO

0–12 h No gross change. **Red neurons** from ~12 h.

12–24 h Pale, soft, swollen; **grey–white junction blurred**.

1–3 d Neutrophils.

3–7 d Macrophages; **liquefactive necrosis** begins.

1–2 wk Peak macrophages; reactive **gliosis** at the rim.

> 3 wk **Cystic cavity with a gliotic wall** — no collagen scar.

Figure 2. Horizontal section. One hemisphere shows brown discoloration, softening and early cavitation in the **MCA territory** with loss of the grey–white junction; the other is normal.

Table 2. *The four ischemic subtypes*

Subtype	Mechanism	Gross	Clue
Thrombotic	Plaque at ICA bifurcation, MCA origin	Pale	Stuttering onset, prior TIA
Embolic	Cardiac (AF) or artery-to-artery	Hemorrhagic	Maximal at onset; MCA
Lacunar	Lipohyalinosis of penetrators (HTN)	< 15 mm cavity	No cortical signs
Watershed	Global hypoperfusion	Border-zone wedge	“Man in a barrel”

IDENTIFY: Horizontal section of the cerebral hemispheres. One hemisphere contains a sharply demarcated zone of brown discoloration with tissue softening and **early cavitation** involving cortex, subcortical white matter and deep grey, in the distribution of the **middle cerebral artery**; the **grey–white junction is lost** and the adjacent ventricle is distorted. The opposite hemisphere is preserved for comparison.

MORPHOLOGY: Evolution is time-stamped (left panel). The end state is a **fluid-filled cyst lined by reactive astrocytes** — the CNS heals by **gliosis, not fibrosis**, so there is no collagen scar.

DISEASE: Ischemic cerebral infarction. The CNS is the one organ where infarction is **liquefactive, not coagulative**.

ETIOLOGY: Thrombotic — atherosclerosis at branch points. Embolic — **atrial fibrillation**, mural thrombus after MI, infective endocarditis, artery-to-artery from a carotid plaque; the **MCA is the vessel most often involved**. Lacunar — hypertensive small-vessel disease. Global hypoperfusion — watershed. Less common: dissection, vasculitis, hypercoagulable states, sickle cell disease.

PATHOGENESIS: Occlusion → ATP failure → Na^+/K^+ pump failure and cytotoxic edema → **glutamate excitotoxicity**, calcium influx, free radicals → neuronal death. The irreversible **core** is surrounded by a hypoperfused but viable **penumbra** — **the penumbra is what reperfusion is trying to save**.

STRUCTURAL CHANGES: Cytotoxic then vasogenic edema **peaking at 3–5 days** → mass effect, midline shift, subfalcine and uncal herniation. Remote: cystic cavity with a gliotic wall and ex vacuo ventricular dilation.

SIGNS / SYMPTOMS: Sudden focal deficit confined to one vascular territory, **maximal at onset** and afebrile. Cortical signs (aphasia, neglect, hemianopia) mean a large vessel; their **absence with a pure motor or pure sensory deficit means lacunar**.

INVESTIGATIONS: **Non-contrast CT first — to exclude hemorrhage, not to see the infarct**, which may stay normal for 6–24 h (early signs: hyperdense MCA, loss of the insular ribbon, sulcal effacement). **Diffusion-weighted MRI is positive within minutes**. CTA for large-vessel occlusion; then ECG/telemetry for AF, carotid duplex, echocardiography, lipids and HbA1c.

COURSE & COMPLICATIONS: **Thrombolysis within 4.5 h**; thrombectomy for large-vessel occlusion up to 24 h with imaging selection. Hemorrhagic transformation (embolic, reperfused or anticoagulated); malignant MCA edema and herniation at days 3–5; later epilepsy and vascular dementia. Prevention: antiplatelet, statin, blood-pressure control, **anticoagulation for AF → Mural Thrombus → Embolism in Brain**.

Slide 6 — Intracerebral Hemorrhage

A space-occupying clot in the deep grey matter — the site names the cause



SITE → SYNDROME (HYPERTENSIVE)

Putamen ~50–60%. Hemiplegia; **eyes deviate toward the lesion.**

Thalamus Hemisensory loss; eyes forced down and in.

Pons Pinpoint reactive pupils, coma, quadriplegia.

Cerebellum Occipital headache, ataxia, **no hemiparesis** — decompress.

Lobar Not hypertensive — think **amyloid angiopathy.**

Figure 3. Horizontal section. A large clot expands and destroys the **deep grey matter** of one hemisphere, compressing the ventricle, shifting the midline and **entering the ventricles.**

Table 3. *The three-way discrimination*

	Deep (hypertensive)	Lobar (amyloid)	Subarachnoid
Site	Putamen, thalamus, pons, cerebellum	Cortico-subcortical, occipital/parietal	Basal cisterns
Vessel	Lenticulostriate; Charcot–Bouchard	Aβ in cortical vessel walls	Saccular aneurysm , AComm
Patient	Chronic hypertension, 50–70	Elderly, normotensive, recurrent	40–60; ADPKD, smoking
Onset	Deficit worsening over minutes	Focal deficit ± seizure	Thunderclap headache

IDENTIFY: Horizontal section with a large, sharply circumscribed mass of **clotted blood** filling and destroying the deep grey matter of one hemisphere. The clot **displaces rather than infiltrates** the surrounding parenchyma, effaces the ipsilateral ventricle, shifts the midline and ruptures into the ventricular system.

MORPHOLOGY: Acute — clotted blood dissecting along white-matter tracts with perihematoma edema and anoxic neuronal change. Weeks to months — clot breakdown with **hemosiderin-laden macrophages** and a residual **slit-like cavity with a brown gliotic rim.**

DISEASE: Spontaneous (non-traumatic) intracerebral hemorrhage — here the deep, **ganglionic, hypertensive pattern.**

ETIOLOGY: **Hypertension is the leading cause**, at deep sites in descending frequency: **putamen > thalamus > pons > cerebellum.** Other causes are identified by their pattern: **cerebral amyloid angiopathy** → recurrent lobar bleeds in the elderly (Aβ in vessel walls, *Congo red* with apple-green birefringence); AVM or cavernous malformation in the young; anticoagulation and thrombolysis; hemorrhage into a tumour (glioblastoma, melanoma, renal cell); cocaine and sympathomimetics; vasculitis.

PATHOGENESIS: Chronic hypertension → **hyaline arteriosclerosis and lipohyalinosis** of the small penetrating lenticulostriate branches → mural weakening with **Charcot–Bouchard microaneurysms (< 300 μm)** → rupture. The hematoma **expands over minutes to hours**, and secondary injury follows from the clot itself (thrombin, iron) and from perihematoma edema.

STRUCTURAL CHANGES: Space-occupying hematoma with ventricular compression and midline shift; **intraventricular extension** risking obstructive hydrocephalus; transtentorial and uncal herniation with **Duret hemorrhages** in the brainstem.

SIGNS / SYMPTOMS: Abrupt focal deficit that **progresses over minutes to hours** — unlike the embolic infarct, which is maximal at onset — with **headache, vomiting, depressed consciousness** and markedly elevated blood pressure. Site-specific syndromes are in the left panel.

INVESTIGATIONS: **Non-contrast CT is immediately diagnostic:** hyperdense acute blood. CTA “spot sign” predicts expansion. **GRE/SWI MRI microbleeds** distinguish the lobar amyloid distribution from the deep hypertensive one. Coagulation studies, platelets, toxicology; angiography if the patient is young, normotensive or the bleed is lobar (AVM, aneurysm).

COURSE & COMPLICATIONS: High mortality — **roughly 35–50% at 30 days.** Controlled blood-pressure lowering, reversal of anticoagulation, ICP management; **surgical evacuation for cerebellar hemorrhage** or neurological deterioration. Survivors are left with a slit cavity, fixed deficits and a risk of seizures.

A 66-year-old man with 20 years of poorly controlled hypertension develops sudden left-sided weakness while straining. Over the next 45 minutes he becomes progressively drowsy, vomits twice, and his headache worsens. Blood pressure is 218/124 mm Hg. Examination shows a left hemiplegia involving the face, left hemisensory loss, and both eyes deviated to the right. Non-contrast CT shows a hyperdense mass in the right putamen effacing the right lateral ventricle, with 8 mm of midline shift.

Which of the following best describes the vascular lesion responsible?

- A. Atheromatous plaque at the internal carotid bifurcation giving artery-to-artery emboli
- B. Saccular outpouching at the junction of the anterior communicating and anterior cerebral arteries
- C. Charcot–Bouchard microaneurysms of the lenticulostriate branches arising on lipohyalinosis
- D. β -amyloid deposition in cortical and leptomeningeal vessel walls
- E. Fibrinoid necrosis of arterioles following an acute rise in blood pressure

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ANSWER — C. Hypertensive intracerebral hemorrhage

Why: Chronic hypertension makes the small **lenticulostriate penetrators** lipohyalinotic, and the weakened walls form **Charcot–Bouchard microaneurysms** that rupture. The **putamen** is the single most common site. A deficit that **worsens over minutes** with headache, vomiting and a very high pressure is blood, not an infarct — which is maximal at onset. The eyes deviate toward the lesion, away from the hemiplegia.

Distractors: A = embolic infarct (maximal at onset; CT would not be hyperdense). B = AComm aneurysm → subarachnoid hemorrhage with thunderclap headache. D = amyloid angiopathy → lobar bleed in an elderly normotensive. E = malignant hypertension / posterior reversible encephalopathy, not a focal clot.